

# MASTERMIND



Mastermind is a code-breaking game for two players. One player is the codemaker and secretly creates a color code, while the other player is the codebreaker and tries to figure out that code by making guesses and interpreting feedback.

At the start of a round, the codemaker creates a hidden code using four colored pegs. In the standard game, the code is made from a set of six possible colors, and colors may be repeated, so the code can contain two or more pegs of the same color. The code is placed where the codebreaker cannot see it.

The codebreaker then makes a first guess by placing four colored pegs in a row. That guess is meant to match both the colors and the positions of the hidden code. After each guess, the codemaker gives feedback using small key pegs. A black key peg means that one guessed peg has the correct color in the correct position. A white key peg means that one guessed peg has the correct color but is in the wrong position.

The feedback tells how many pegs are correct, but it does not identify which specific guessed pegs earned which feedback pegs. In other words, the codebreaker learns how close the guess was, but not exactly which positions were right or wrong. When repeated colors are involved, each feedback peg applies only once, and exact matches take priority over color-only matches.

After receiving feedback, the codebreaker makes another guess in the next row, and the process repeats. Earlier guesses and their feedback stay visible, so the codebreaker can use them to narrow down the possibilities over time. The round ends when the codebreaker guesses the exact code or runs out of available rows for guesses.

In many standard versions, the board allows ten or twelve guesses, depending on the edition. In a basic win-loss version, the codebreaker wins by correctly guessing the secret code within the available guesses; otherwise the codemaker wins.